

ec2 task

1. Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache.

—> create a instance and before creating it add script for httpd

User data - optional | [Info](#)
Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
yum update -y
yum install -y httpd
systemctl enable httpd
systemctl start httpd
echo "<h1>Amazon Linux 2 - Apache from user-data</h1>" > /var/www/html/index.html
```

☐ User data has already been base64 encoded

—>And launch the instance

Find Instance by attribute or tag (case-sensitive)					All states ▾				
	Name 	Instance ID	Instance state 	Instance type 	Status check	Alarm status	Availability Zone 	Public IPv4 DNS 	Pu
☐	server2	i-0fcda8556b7579b36	 Stopped  	t3.micro	–	View alarms +	eu-north-1c	–	–
☒	server1	i-02ba49321ea4734a1	 Stopped  	t3.micro	–	View alarms +	eu-north-1c	–	–
☐	server-apache	i-0b0f9a616d2c889af	 Running  	t3.micro	 Initializing	View alarms +	eu-north-1b	ec2-16-171-162-252.eu...	16

—> We have created a server-apache named server

—> now open the server in local terminal

[illegible]

```
waseemakram@waseem ~ % cd downloads
waseemakram@waseem downloads % ssh -i "waseem.pem" ec2-user@ec2-16-171-162-252.eu-north-1.compute.amazonaws.com
```

```
~_#####_ Amazon Linux 2023
~_ \#####\
~_ \###|
~_ \#/ ____ https://aws.amazon.com/linux/amazon-linux-2023
~_ V-' '->
~_ /
~_ /
~_ /m/'
```

```
[ec2-user@ip-172-31-36-88 ~]$ systemctl status httpd
```

```
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Wed 2025-11-19 09:54:23 UTC; 1min 36s ago
     Docs: man:httpd.service(8)
 Main PID: 3276 (httpd)
    Status: "Total requests: 2; Idle/Busy workers 100/0;Requests/sec: 0.0225; Bytes served/sec: 17 B/sec"
   Tasks: 177 (limit: 1053)
  Memory: 13.5M
      CPU: 140ms
    CGroup: /system.slice/httpd.service
            └─3276 /usr/sbin/httpd -DFOREGROUND
              ├─3421 /usr/sbin/httpd -DFOREGROUND
              ├─3422 /usr/sbin/httpd -DFOREGROUND
              ├─3423 /usr/sbin/httpd -DFOREGROUND
              └─3424 /usr/sbin/httpd -DFOREGROUND
```

```
Nov 19 09:54:23 ip-172-31-36-88.eu-north-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Nov 19 09:54:23 ip-172-31-36-88.eu-north-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Nov 19 09:54:23 ip-172-31-36-88.eu-north-1.compute.internal httpd[3276]: Server configured, listening on: port 80
[ec2-user@ip-172-31-36-88 ~]$
```

→ httpd is already active as we can see in image



2) Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.

—> select a AMI of ubuntu

Name and tags [Info](#)

Name

ubuntu-nginx

Add additional tags

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

Q

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

Free tier eligible

—> give instance name ubuntu-nginx for understanding

—> add the bash script for installing nginx

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

↑ Choose file

```
#!/bin/bash
apt update -y
apt install nginx -y
systemctl start nginx
systemctl enable nginx
echo "nginx started..."
```

☐ User data has already been base64 encoded

—> and create that instance

☐	server-apache	i-0b0f9a616d2c889af	Running	t3.micro	3/3 checks passed	View alarms +	eu-north-1b	ec2-16-171-
☒	ubuntu-nginx	i-0f15ea4c280f410f6	Running	t3.micro	Initializing	View alarms +	eu-north-1b	ec2-51-21-1

—> now add the server in local

—> we can see nginx is already installed and running

```
waseemakram@waseem downloads % ssh -i "waseem.pem" ubuntu@ec2-51-21-152-67.eu-north-1.compute.amazonaws.com
The authenticity of host 'ec2-51-21-152-67.eu-north-1.compute.amazonaws.com (51.21.152.67)' can't be established.
ED25519 key fingerprint is SHA256:54dU8prwTcvkVQEIn0xTmuMxV0S15bIcd3vnXtq8CEE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-51-21-152-67.eu-north-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Wed Nov 19 10:20:29 UTC 2025

System load:  0.75          Temperature:   -273.1 C
Usage of /:   30.4% of 6.71GB Processes:      131
Memory usage: 39%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.42.198

Expanded Security Maintenance for Applications is not enabled.

20 updates can be applied immediately.
19 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-42-198:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-11-19 10:20:26 UTC; 37s ago
     Docs: man:nginx(8)
   Main PID: 1553 (nginx)
    Tasks: 3 (limit: 1008)
   Memory: 2.4M (peak: 5.3M)
      CPU: 28ms
   CGroup: /system.slice/nginx.service
           └─1553 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─1555 "nginx: worker process"
               └─1556 "nginx: worker process"

Nov 19 10:20:26 ip-172-31-42-198 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Nov 19 10:20:26 ip-172-31-42-198 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-42-198:~$
```


Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

3) Launch one Windows server and install Tomcat on Windows.

Name and tags [Info](#)

Name

window-tomcat

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or click [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

[Quick Start](#)

 aws	 Mac	 ubuntu®	 Microsoft	 Red Hat	 SUSE	 debian
--	--	--	--	--	---	--


[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace
and the Community

Amazon Machine Image (AMI)

Microsoft Windows Server 2025 Base
ami-010e40c6557403885 (64-bit (x86))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Create a server with. Ami window

—> now after launching instance

—> click connect and open rep client and click set password

Now add your key and decrypt and copy the password and open the remote desktop file

Connect to an instance using the browser-based client.

Session Manager

RDP client

EC2 serial console

Record RDP connections

You can now record RDP connections using AWS Systems Manager just-in-time node access. [Learn more](#)

Instance ID

i-002fc1ecd4154a5d0 (window-tomcat)

Connection Type

Connect using RDP client

Download a file to use with your RDP client and retrieve your password.

Connect using Fleet Manager

To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed on the instance. For more information, see [Working with SSM Agent](#)

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

Download remote desktop file

When prompted, connect to your instance using the following username and password:

Public DNS

ec2-13-60-173-172.eu-north-1.compute.amazonaws.com

Username

Info

Administrator

Password

(CSdSo%0eHzb@HqN?niPbRrMskdWmmOd

If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

key pair associated with this instance

window-key

Private key

Either upload your private key file or copy and paste its contents into the field below.

Upload private key file

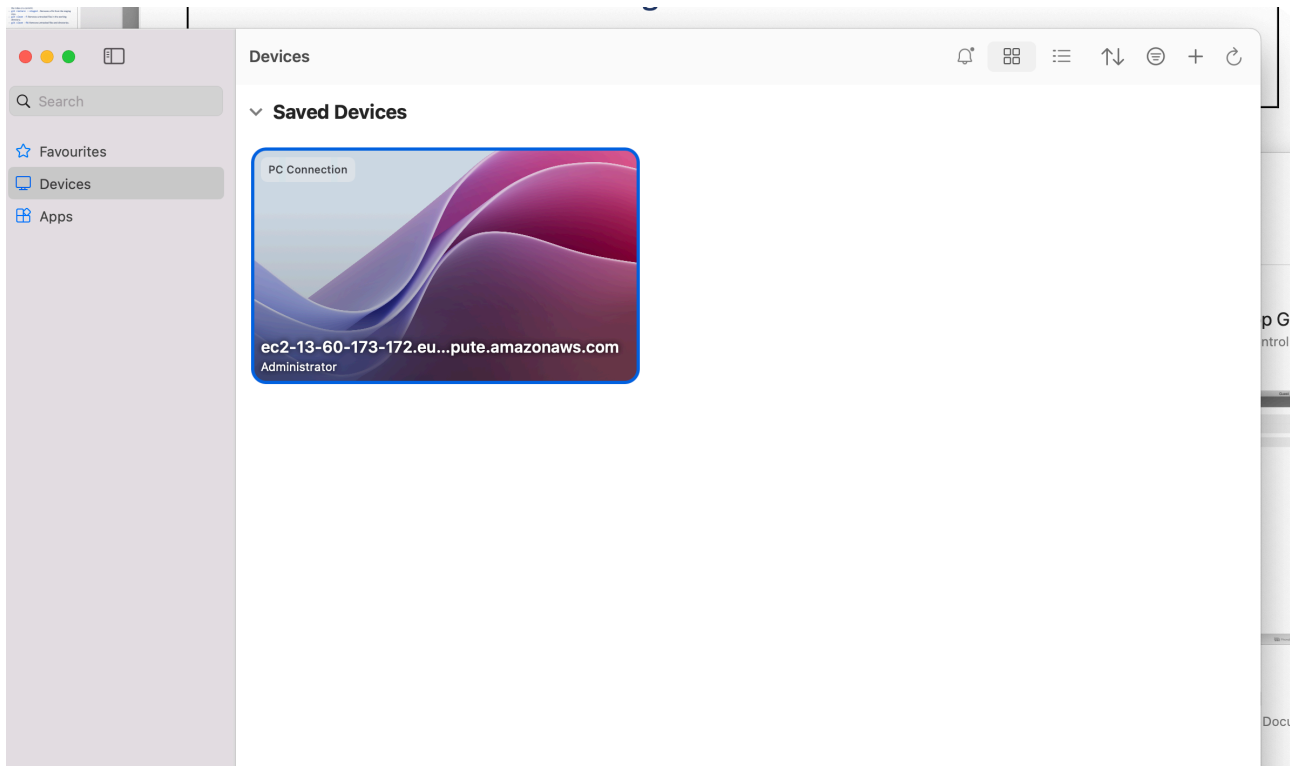
window-key.pem

X

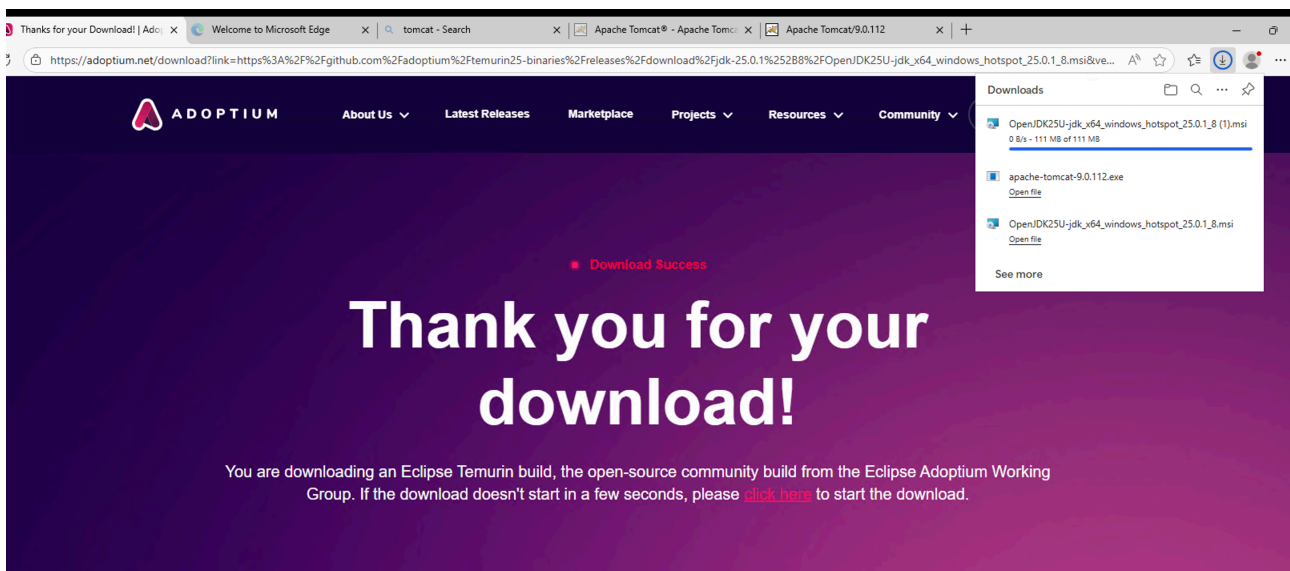
1.67 KB

Private key contents

```
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEAyVdoOjKb4TimV9VyLvBjv7K1W03+PPgl2MBkQg8AfG9MBIjy
wdS4qJrMRPfv3DOFwN0tFuWmKuoCKOXbj9I7mJ0m7jGSZQOPex4aXyYFRiLieh9W
GDvoYWUHGyi2qhErDzZDYFQ8xu8tCZPT4mz9iad4pruNgD1WJYM1SnrVqTTjefeg
L2XrKIHR1ls+QDXSiLX1mOcnXaovITC4jfbJH2cWh4inztpVv3ab89UX4bXV0Z+
NwMNVfQec+JvlxgSCv4LWYXfxn1z95QR4HZk7CdcqJ49KKhEwfyF/0ZrEdMcAth
Y4+EIFZ9G6SwQpK423XI7GFeFBV2OVgSGB2cpQIDAQABAoIBABuBVlrE+SiObaMI
JL4m1umkR0Q+5mF74Oe4eLnRhNZ6+d7QHfG+1q8p3of5ewoQ/08EVLmR67GZc6wU
-----
```



Open the window



—>And download all applications in window

Welcome to Microsoft Edge

tomcat - Search

Apache Tomcat® - Apache Tomcat

Apache Tomcat/9.0.112

Home

Documentation

Configuration

Examples

Wiki

Mailing Lists

Find Help

Apache Tomcat/9.0.112



If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

[Security Considerations How-To](#)

[Manager Application How-To](#)

[Clustering/Session Replication How-To](#)

Server Status

Manager App

Host Manager

Developer Quick Start

[Tomcat Setup](#)

[First Web Application](#)

[Realms & AAA](#)

[JDBC Data Sources](#)

[Examples](#)

[Servlet Specifications](#)

[Tomcat Versions](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:
`$CATALINA_HOME/conf/tomcat-users.xml`
In Tomcat 9.0 access to the manager application is split between different users.
[Read more...](#)

[Release Notes](#)

[Changelog](#)

[Migration Guide](#)

[Security Notices](#)

Documentation

[Tomcat 9.0 Documentation](#)

[Tomcat 9.0 Configuration](#)

[Tomcat Wiki](#)

Find additional important configuration information in:
`$CATALINA_HOME/RUNNING.txt`
Developers may be interested in:

[Tomcat 9.0 Bug Database](#)

[Tomcat 9.0 JavaDocs](#)

[Tomcat 9.0 Git Repository at GitHub](#)

Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:

[tomcat-announce](#)

Important announcements, releases, security vulnerability notifications. (Low volume).

[tomcat-users](#)

User support and discussion

[taglibs-user](#)

User support and discussion for [Apache Taglibs](#)

[tomcat-dev](#)

Development mailing list, including commit messages

Other Downloads

[Tomcat Connectors](#)

[Tomcat Native](#)

[Taglibs](#)

[Deployer](#)

Other Documentation

[Tomcat Connectors](#)

[mod_jk Documentation](#)

[Tomcat Native](#)

[Deployer](#)

Get Involved

[Overview](#)

[Source Repositories](#)

[Mailing Lists](#)

[Wiki](#)

Miscellaneous

[Contact](#)

[Legal](#)

[Sponsorship](#)

[Thanks](#)

Apache Software Foundation

[Who We Are](#)

[Heritage](#)

[Apache Home](#)

[Resources](#)

Copyright ©1999-2025 Apache Software Foundation. All Rights Reserved



http://ec2-13-60-173-172.eu-north-1.compute.amazonaws.com:8080

Not secure ec2-13-60-173-172.eu-north-1.compute.amazonaws.com:8080

Home

Documentation

Configuration

Examples

Wiki

Mailing Lists

Find Help

Apache Tomcat/9.0.112



If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

[Security Considerations How-To](#)

[Manager Application How-To](#)

[Clustering/Session Replication How-To](#)

Server Status

Manager App

Host Manager

Developer Quick Start

[Tomcat Setup](#)

[First Web Application](#)

[Realms & AAA](#)

[JDBC Data Sources](#)

[Examples](#)

[Servlet Specifications](#)

[Tomcat Versions](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:
`$CATALINA_HOME/conf/tomcat-users.xml`
In Tomcat 9.0 access to the manager application is split between different users.
[Read more...](#)

[Release Notes](#)

[Changelog](#)

[Migration Guide](#)

[Security Notices](#)

Documentation

[Tomcat 9.0 Documentation](#)

[Tomcat 9.0 Configuration](#)

[Tomcat Wiki](#)

Find additional important configuration information in:
`$CATALINA_HOME/RUNNING.txt`
Developers may be interested in:

[Tomcat 9.0 Bug Database](#)

[Tomcat 9.0 JavaDocs](#)

[Tomcat 9.0 Git Repository at GitHub](#)

Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:

[tomcat-announce](#)

Important announcements, releases, security vulnerability notifications. (Low volume).

[tomcat-users](#)

User support and discussion

[taglibs-user](#)

User support and discussion for [Apache Taglibs](#)

[tomcat-dev](#)

Development mailing list, including commit messages

4) Take a snapshot of the instance created in Task 1.

What is a Snapshot in AWS?

A **Snapshot** is a **backup** of an EBS Volume
(where your EC2 instance's storage / disk lives).

Snapshot = **Photo / Backup of your server's disk**

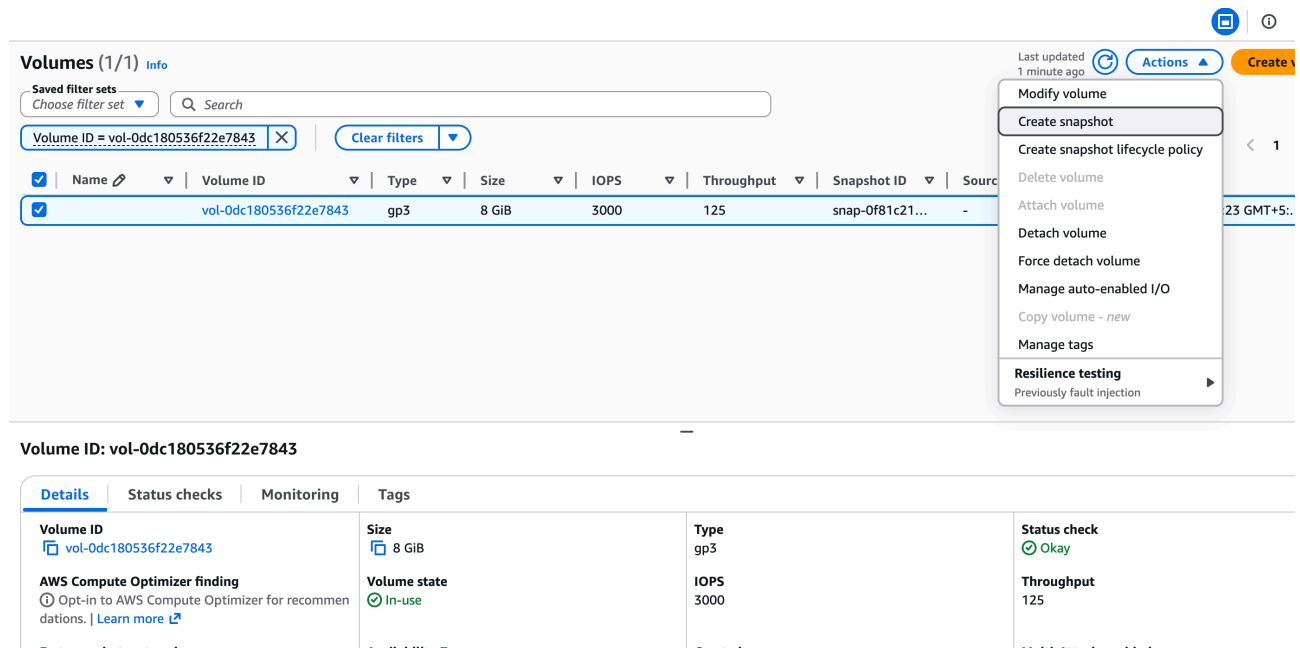
It contains:

- Operating system (Amazon Linux)
- Apache installation
- Your user-data installed files
- Any files you had on the server

So if your instance is deleted or corrupted, you can **restore** a new instance using the snapshot.

—> click the instance and go to volumes

—> in actions choose create snapshot



The screenshot displays the AWS Management Console's 'Volumes' page. At the top, there's a search bar and a filter set. Below, a table lists volumes. The first volume, 'vol-0dc180536f22e7843', is selected. An 'Actions' dropdown menu is open, showing options like 'Modify volume', 'Create snapshot', 'Delete volume', 'Attach volume', 'Detach volume', 'Force detach volume', 'Manage auto-enabled I/O', 'Copy volume - new', 'Manage tags', and 'Resilience testing'. The 'Create snapshot' option is highlighted. Below the table, the 'Details' tab for the selected volume is shown, displaying its ID, size (8 GiB), type (gp3), and status (In-use). Other details include 'AWS Compute Optimizer finding' and 'Status check' (Okay).

Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source
vol-0dc180536f22e7843	gp3	8 GiB	3000	125	snap-0f81c21...	-

Volume ID: vol-0dc180536f22e7843

Details	Status checks	Monitoring	Tags
Volume ID vol-0dc180536f22e7843 AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more	Size 8 GiB Volume state In-use	Type gp3 IOPS 3000	Status check Okay Throughput 125

And add discription to the snapshot
And create it

✔ Successfully created snapshot `snap-0d8a9adb374c0040f` from volume `vol-0dc180536f22e7843`.
If you need your snapshot to be immediately available consider using Fast Snapshot Restore.

Volumes (1) Info

Saved filter sets
Choose filter set

Search

Volume ID = `vol-0dc180536f22e7843` Clear filters

	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source
☐		<code>vol-0dc180536f22e7843</code>	gp3	8 GiB	3000	125	<code>snap-0f81c21...</code>	-

—>Now check out the snapshot list we can find the snapshot of task 1

Snapshots (1/1) Info

Last updated 1 minute ago Recycle Bin Actions Create snapshot

Owned by me

Search

☒	Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot status	Started
☒		<code>snap-0d8a9adb374c0040f</code>	1.68 GiB	8 GiB	snapshot of amazon linux ...	Standard	Completed	2025/11/19 19:07 GMT+5:

Snapshot ID: `snap-0d8a9adb374c0040f`

Details

Snapshot settings

Storage tier

Tags

Snapshot ID

`snap-0d8a9adb374c0040f`

Owner

`207662791773`

Description

Full snapshot size

`1.68 GiB`

Started

Wed Nov 19 2025 19:07:40 GMT+0530 (India Standard Time)

Progress

100%

Product codes

-

Snapshot status

Completed

Fast snapshot restore

-

5) Assign passwordless authentication for the EC2 created in Task 2.

```
ubuntu@ip-172-31-42-198:~$ ssh-keygen -t rsa
Generating public/private rsa key pair.
[Enter file in which to save the key (/home/ubuntu/.ssh/id_rsa): /home/ubuntu/.ssh/id_rsa
[Enter passphrase (empty for no passphrase):
[Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_rsa
Your public key has been saved in /home/ubuntu/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:8LqnqUcqpHYaup4E0y2MuEZpfrwiaXM3YMMqjHwjzWI ubuntu@ip-172-31-42-198
The key's randomart image is:
+---[RSA 3072]-----+
|
| .
| .
| .
|+. . .
|+= o . S
|=. *.+
|B*===
|@E=Ooo
|X*@oo .
+---[SHA256]-----+
ubuntu@ip-172-31-42-198:~$ cat /home/ubuntu/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDQFQRTZbnsjJ9MCUD/4M8nzc04psWiT3JArlOnLQtPVBj73rGgbNmmNmK2TgDl3xqi24wMqBad3KBISux0T+z8I
HGx7z9q5cmEF0gkiliuEZcd48/WxXelMhlr08acPhlC2aV1LY0hxh2viahABKmz80ukScbD2XD1BbTso6AV6HedCchrFVeKB8IgD1/8J6MWDblQp4z+e8kcRCm1zLw
-31-42-198
```

Open the ubuntu server

—> now create a ssh-keygen.

And copy the

—> .ssh/id_rsa.pub

And copy it

—> On the Ubuntu server, created the .ssh directory

—> Opened the authorized_keys file: vi ~/.ssh/authorized_keys

```
ubuntu@ip-172-31-42-198:~$ mkdir -p ~/.ssh
chmod 700 ~/.ssh
ubuntu@ip-172-31-42-198:~$ vi ~/.ssh/authorized_keys
ubuntu@ip-172-31-42-198:~$ vi ~/.ssh/authorized_keys
ubuntu@ip-172-31-42-198:~$ chmod 700 ~/.ssh
ubuntu@ip-172-31-42-198:~$ chmod 600 ~/.ssh/authorized_keys
ubuntu@ip-172-31-42-198:~$ vi ~/.ssh/authorized_keys
ubuntu@ip-172-31-42-198:~$
```

Now paste the key below of that key

(This key is allowed to connect without password.)

```
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDQFQRTZbnsjJ9MCUD/4M8nzc04psWiT3JArlOnLQtPVBj73rGgbNmmNmK2TgDl3xqi24wMqBad3KBISux0T+z8I
HGx7z9q5cmEF0gkiliuEZcd48/WxXelMhlr08acPhlC2aV1LY0hxh2viahABKmz80ukScbD2XD1BbTso6AV6HedCchrFVeKB8IgD1/8J6MWDblQp4z+e8kcRCm1zLw
-31-42-198
```

—> Tested the SSH login from the Mac:

```
Connection to ec2-16-170-233-124.eu-north-1.compute.amazonaws.com closed.
[waseemakram@waseem downloads % ssh -i "waseem.pem" ubuntu@ec2-16-170-233-124.eu-north-1.compute.amazonaws.com
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Nov 19 13:59:20 UTC 2025

System load:  0.0           Temperature:   -273.1 C
Usage of /:   30.5% of 6.71GB Processes:      113
Memory usage: 23%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.42.198

Expanded Security Maintenance for Applications is not enabled.

20 updates can be applied immediately.
19 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Nov 19 13:49:17 2025 from 14.192.14.62
ubuntu@ip-172-31-42-198:~$
```

—>The login succeeded **without any password prompt**, confirming that passwordless authentication was correctly set

```
ubuntu@ip-172-31-42-198:~$ cd /home/ubuntu/.ssh
ubuntu@ip-172-31-42-198:~/.ssh$ ls
authorized_keys  id_rsa  id_rsa.pub
ubuntu@ip-172-31-42-198:~/.ssh$
```

Passwordless SSH authentication was successfully enabled for the Ubuntu EC2 instance. Now the user can log in securely using their SSH key **without entering a password**.

6)Launch Any EC2 Using the Spot Purchasing Option

Before creating instance

—> add additional details

In side additional details

Scroll down and add spot instances

Purchasing option | [Info](#)
☐ None
☐ Capacity Blocks
Launch instances for your active capacity blocks
☒ Spot instances
Request Spot Instances at the Spot price, capped at the On-Demand price

[Discard Spot instance options](#)

Spot Instance Options | [Info](#)
Specify Spot Instance Options such as Maximum Price, Request type, expiration date and interruption behavior
Maximum price | [Info](#)
☒ No maximum price
Request Spot Instances at the Spot price, capped at the On-Demand price
☐ Set your maximum price (per instance/hour)

Request type | [Info](#)

Select ▼

Valid to | [Info](#)
☒ No request expiry date
The default value is no expiry date
☐ Set your request expiry date

Interruption behavior | [Info](#)

Terminate ▼

—>Change request type

—>change interruption behaviour

And later create the instance

—> Hence we launched a ec2 with spot options

Monitoring
disabled

Allowed image
-

Launch time
📅 Wed Nov 19 2025 20:05:55 GMT+0530 (India Standard Time) (9 minutes)

Instance auto-recovery
Default

AMI Launch index

Platform details
📁 Linux/UNIX

Termination protection
Disabled

AMI location
📁 amazon/al2023-ami-2023.9.20251117.1-kernel-6.1-x86_64

Lifecycle
spot

Key pair assigned at launch

7) Enable termination policy on the EC2 created in Task 2.

The screenshot shows the AWS Management Console interface. At the top, there's a navigation bar with the account name 'Europe (Stockholm)' and 'Account ID: 2076...'. Below this, a green banner indicates 'Successfully initiated starting of i-0f15ea4c280f410f6'. The main section is titled 'Instances (1/6) Info'. A table lists several instances, with 'ubuntu-nginx' (ID: i-0f15ea4c280f410f6) highlighted in blue. The 'Actions' dropdown menu is open, showing options like 'Attach to Auto Scaling Group', 'Change termination protection', 'Change stop protection', etc. The 'Change termination protection' option is highlighted. Below the table, the details for the selected instance 'i-0f15ea4c280f410f6 (ubuntu-nginx)' are shown, including its state (Running) and public IP address (13.53.147.234).

Name	Instance ID	Instance state	Instance type	Status check
server2	i-0fcda8556b7579b36	Stopped	t3.micro	-
server1	i-02ba49321ea4734a1	Stopped	t3.micro	-
server-apache	i-0b0f9a616d2c889af	Stopped	t3.micro	-
ubuntu-nginx	i-0f15ea4c280f410f6	Running	t3.micro	Initializing
window-tomcat	i-002fc1ecd4154a5d0	Stopped	t3.micro	-
spot-instance	i-097f086c6c4ce1d83	Running	t3.micro	3/3 checks passed

Click

Actions

→ Instance settings → Change termination protection

The screenshot shows the 'Change termination (deletion) protection' dialog box. The dialog has a title bar with a close button. The main text says: 'To prevent your instance from being accidentally deleted, you can enable termination protection for the instance. [Learn more](#)'. Below this, the 'Instance ID' is 'i-0f15ea4c280f410f6 (ubuntu-nginx)'. The 'Termination protection' section shows a checkbox labeled 'Enable' which is checked. At the bottom, there are 'Cancel' and 'Save' buttons. The background shows the same EC2 Instances page as the previous screenshot, but it is dimmed.

Now your EC2 instance **cannot be terminated accidentally**.

If anyone tries to terminate it, AWS will show an error

8)Launch one EC2 using AWS CLI.

—>Installed awscli in local terminal

```
waseemakram@waseem downloads % brew install awscli
=> Auto-updating Homebrew...
Adjust how often this is run with `HOMEBREW_AUTO_UPDATE_SECS` or disable with
`HOMEBREW_NO_AUTO_UPDATE=1`. Hide these hints with `HOMEBREW_NO_ENV_HINTS=1` (see `man brew`).
=> Downloading https://ghcr.io/v2/homebrew/core/portable-ruby/blobs/sha256:c6946ba2c387b47934e77c352c2056489421003ec7ddb2abf246cef2168ec140
#####
=> Pouring portable-ruby-3.4.7.arm64_big_sur.bottle.tar.gz
=> Auto-updated Homebrew!
Updated 3 taps (hashicorp/tap, homebrew/core and homebrew/cask).
=> New Formulae
aklomp-base64: Fast Base64 stream encoder/decoder in C99, with SIMD acceleration
bulletty: Pretty feed reader (ATOM/RSS) that stores articles in Markdown files
claude-code-templates: CLI tool for configuring and monitoring Claude Code
container: Create and run Linux containers using lightweight virtual machines
corepack: Package acting as bridge between Node projects and their package managers
docmd: Minimal Markdown documentation generator
doh: Stand-alone DNS-over-HTTPS resolver using libcurl
fake-gcs-server: Emulator for Google Cloud Storage API
forgejo-cli: CLI tool for interacting with Forgejo
gitnr: Create `.gitignore` using templates from TopTal, GitHub or your own collection
goclone: Website Cloner
gotun: Lightweight HTTP proxy over SSH
gruyere: TUI program for viewing and killing processes listening on ports
gwctl: CLI for managing and inspecting Gateway API resources in Kubernetes clusters
hapless: Run and manage background processes
helm@3: Kubernetes package manager
hexhog: Hex viewer/editor
hf-mcp-server: MCP Server for Hugging Face
icu4c@78: C/C++ and Java libraries for Unicode and globalization
intelli-shell: Like IntelliSense, but for shells
kagent: Kubernetes native framework for building AI agents
kanata-tray: System tray for kanata keyboard remapper
ktea: Kafka TUI client
libaegis: Portable C implementations of the AEGIS family of encryption algorithms
libdecor: Client-side decorations library for Wayland client
```

—>Created a access key

Access keys (1)

Actions ▼ Create access key

Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. [Learn more](#)

	Access key ID	Created on	Access key last used	Region last used	Service last used	Status
☐	AKIATAWNKXROQTKQ5PVR	9 minutes ago	None	N/A	N/A	✔ Active

CloudFront key pairs (0)

Actions ▼ Upload Create CloudFront key pair

You use key pairs in Amazon CloudFront to create signed URLs. You can have a maximum of two CloudFront key pairs (active or inactive) at a time.

Now

—> used aws configure command for configuring key and password from above access key and added json as default output

```
waseemakram@waseem downloads % aws configure

AWS Access Key ID [*****2KNH]: AKIATAWNKXROQTKQ5PVR
AWS Secret Access Key [*****z14X]: 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
Default region name [eu-north-1]:
Default output format [json]:
waseemakram@waseem downloads % aws ec2 run-instances \
```

—> After setting the key and password enter

```
waseemakram@waseem downloads % aws ec2 run-instances \
--image-id ami-07c8c1b18ca66bb07 \
[ --instance-type t3.micro \
--key-name window-key \
--count 1 \
--region eu-north-1

{
  "ReservationId": "r-0a1eb68683c19a08a",
  "OwnerId": "207662791773",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "37d45b99-17ea-4e46-be98-c66aab3f0a20",
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "NetworkInterfaces": [
        {
          "Attachment": {
            "AttachTime": "2025-11-19T18:52:23+00:00",
            "AttachmentId": "eni-attach-009a009ef9463affe",
            "DeleteOnTermination": true,
            "DeviceIndex": 0,
            "Status": "attaching",
            "NetworkCardIndex": 0
          },
          "Description": "",
          "Groups": [
            {
              "GroupId": "sg-0853619dc44ef762e",
              "GroupName": "default"
            }
          ]
        }
      ]
    }
  ]
}
```

```
{
  "Architecture": "x86_64",
  "BlockDeviceMappings": [],
  "ClientToken": "37d45b99-17ea-4e46-be98-c66aab3f0a20",
  "EbsOptimized": false,
  "EnaSupport": true,
  "Hypervisor": "xen",
  "NetworkInterfaces": [
    {
      "Attachment": {
        "AttachTime": "2025-11-19T18:52:23+00:00",
        "AttachmentId": "eni-attach-009a009ef9463affe",
        "DeleteOnTermination": true,
        "DeviceIndex": 0,
        "Status": "attaching",
        "NetworkCardIndex": 0
      },
      "Description": "",
      "Groups": [
        {
          "GroupId": "sg-0853619dc44ef762e",
          "GroupName": "default"
        }
      ],
      "Ipv6Addresses": [],
      "MacAddress": "0a:64:bc:22:1b:d7",
      "NetworkInterfaceId": "eni-0f236df1698d6f88d",
      "OwnerId": "207662791773",
      "PrivateDnsName": "ip-172-31-38-72.eu-north-1.compute.internal",
      "PrivateIpAddress": "172.31.38.72",
      "PrivateIpAddresses": [
        {
          "Primary": true,
          "PrivateDnsName": "ip-172-31-38-72.eu-north-1.compute.internal",
          "PrivateIpAddress": "172.31.38.72"
        }
      ],
      "SourceDestCheck": true,
      "Status": "in-use",
      "SubnetId": "subnet-05f27e3702839a75e",
      "VpcId": "vpc-01126d967b1bb73af",
      "InterfaceType": "interface",
      "Operator": {
        "Managed": false
      }
    }
  ],
  "RootDeviceName": "/dev/sda1",
  "RootDeviceType": "ebs",
  "SecurityGroups": [
    {
      "GroupId": "sg-0853619dc44ef762e",
      "GroupName": "default"
    }
  ],
  "SourceDestCheck": true,
  "StateReason": {
    "Code": "pending",
    "Message": "pending"
  },
  "VirtualizationType": "hvm",
  "CpuOptions": {
    "CoreCount": 1,
    "ThreadsPerCore": 2
  },
  "CapacityReservationSpecification": {
    "CapacityReservationPreference": "open"
  },
  "MetadataOptions": {
```

We have created a ec2 server though our terminal

window-tomcat	i-00z7c1ec04134a20u	Stopped	t3.micro	view alarms	eu-north-1a	
☒	i-0527d1c170f3f2e12	Running	t3.micro	3/3 checks passed View alarms	eu-north-1b	ec2-13-61-10-204.eu-n... 13.61.10.2

i-0527d1c170f3f2e12

